

A New Light-Powered Material Makes Hydrogen From Water

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Chemists at Oregon State University found a new way to make **hydrogen** from water using light. Kyriakos Stylianou led the work. His team built a material called a **photocatalyst**. It can produce **hydrogen** quickly and efficiently.

Hydrogen is used in **fuel cells** for vehicles. It is also used to make ammonia, refine metals and manufacture plastics.

A **catalyst** speeds up a chemical reaction without being permanently changed. A **photocatalyst** is switched on by light. After it absorbs light, it can use that energy to drive reactions faster.

The material is called BVR-19. It has an unusual sulfur-to-sulfur bond that briefly breaks in light. Stylianou said it does not need an extra expensive metal **catalyst**. It also forms on its own in water at room temperature.

Most **hydrogen** today is made from **natural gas**, which releases **carbon dioxide**. It costs about \$1.50 per kilogram, while "green" **hydrogen** costs roughly \$5 per kilogram. Stylianou said the work gives new design rules that could help lower the cost of green **hydrogen**.

Word Treasure

hydrogen -- A light gas that can be used as a clean fuel.

photocatalyst -- A material that uses light to speed up a chemical reaction.

catalyst -- A substance that speeds up a reaction without being permanently changed.

fuel cells -- Devices that turn fuel into electricity.

natural gas -- A fuel that comes from underground.

carbon dioxide -- A gas that is released when many fuels are made or burned.

Break it down

Opening: Oregon State chemists make hydrogen from water with light

What hydrogen is used for

What a catalyst and photocatalyst do

The BVR-19 material and its advantages

Costs and why the work could matter

Why it matters

Cheaper green hydrogen could help reduce the greenhouse gases that come from making hydrogen the usual way.

Test what you remember

1. What does the new material make from water?

- ☐ (A) Hydrogen
- ☐ (B) Plastic
- ☐ (C) Oxygen only
- ☐ (D) Natural gas

2. What switches on a photocatalyst?

- ☐ (A) Light
- ☐ (B) Sound
- ☐ (C) Magnets
- ☐ (D) Heat from a fire

3. What is the material called?

- ☐ (A) BVR-19
- ☐ (B) MaD-5
- ☐ (C) H2O-7
- ☐ (D) OSU-100

4. What does the material not need?

- ☐ (A) Water
- ☐ (B) An extra expensive metal catalyst
- ☐ (C) A lab at Oregon State
- ☐ (D) Light

5. Where does the material form on its own?

- ☐ (A) In space
- ☐ (B) In a hot oven
- ☐ (C) In water at room temperature
- ☐ (D) In frozen air

6. How much does green hydrogen cost, roughly?

- ☐ (A) \$5 per kilogram
- ☐ (B) \$1.50 per kilogram
- ☐ (C) \$50 per kilogram
- ☐ (D) Nothing at all

Answer key (try the quiz first!): 1: A 2: A 3: A 4: B 5: C 6: A

MY THOUGHTS (at least 20 words):